

$$= \frac{1}{2} \lim_{b \rightarrow \infty} [\log(1+x^2)]_0^b$$

$\therefore E(X)$  does not exist.

Ex: Let  $f_X(x) = \begin{cases} \frac{x}{2}, & 0 \leq x \leq 1 \\ \frac{1}{2}, & 1 < x \leq 2 \\ \frac{3-x}{2}, & 2 < x \leq 3 \\ 0, & \text{otherwise} \end{cases}$

Check whether it is a probability density function of a RV  $X$ .  
If so, find the cumulative distribution function, and find  $E(X)$ .

$$\begin{aligned} \int_{-\infty}^{\infty} f(x) dx &= \int_0^1 \frac{x}{2} dx + \int_1^2 \frac{1}{2} dx + \int_2^3 \frac{3-x}{2} dx \\ &= \left[ \frac{x^2}{4} \right]_0^1 + \left[ \frac{x}{2} \right]_1^2 + \left[ -\frac{(3-x)^2}{4} \right]_2^3 \\ &= \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1 \checkmark \end{aligned}$$

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x f_X(x) dx \\ &= \int_0^1 x \cdot \frac{x}{2} dx + \int_1^2 x \cdot \frac{1}{2} dx \\ &\quad + \int_2^3 x \cdot \frac{3-x}{2} dx \end{aligned}$$

The CDF of a RV  $X$  is  $F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt =$

If  $x < 0$ ,  $F_X(x) = \int_{-\infty}^x (0) dx = 0$

If  $0 < x \leq 1$ ,  $F_X(x) = \int_0^x \frac{t}{2} dt = \frac{x^2}{4}$

$1 < x < 2$ ,  $F_X(x) = \int_0^1 \frac{t}{2} dt + \int_1^x \frac{1}{2} dt = \frac{1}{4} + \frac{x-1}{2}$

$$\begin{aligned} 2 < x < 3, \quad F_X(x) &= \int_0^1 \frac{t}{2} dt + \int_1^2 \frac{1}{2} dt + \int_2^x \frac{3-t}{2} dt \\ &= \frac{3}{4} + \left[ -\frac{(3-t)^2}{4} \right]_2^x = \frac{3}{4} + \frac{1}{4} - \frac{(3-x)^2}{4} \\ &= 1 - \frac{(3-x)^2}{4} \end{aligned}$$

$x \geq 3$ ,  $F_X(x) = \int_{-\infty}^x f_X(t) dt = \int_0^1 + \int_1^2 + \int_2^3 = 1$